## Introductory activity

**Activity 1 · Measuring one litre with two cans** 35 min. We have a 9-litre can, a 7-litre can, both without markings, and a water supply. We may fill a can, empty it, or pour one into the other until one is empty or the other is full.

1. Show that one can get exactly 2 litres in the large can.
2. If we pour in $u$ times 9 litres altogether and throw away $v$ times 7 litres, the quantity left is $9u - 7v$. Find natural numbers $u$ and $v$ such that $9u - 7v = 1$. Deduce how to get 1 litre.
3. The cans are replaced by cans of 6 and 9 litres. Show that every quantity of the form $6u + 9v$, with $u$ and $v$ integers, is a multiple of 3. Can one get 1 litre?
4. With cans of $a$ and $b$ litres, which whole quantities does it seem possible to get? State a conjecture involving the greatest common divisor of $a$ and $b$.